

## Data Converters

### Exercise 08 – Delta-Sigma ADC

#### Task I – Analysis of a Second-order Delta-Sigma ADC

The block diagram of a second-order discrete-time Delta-Sigma ADC is shown in Figure 1.

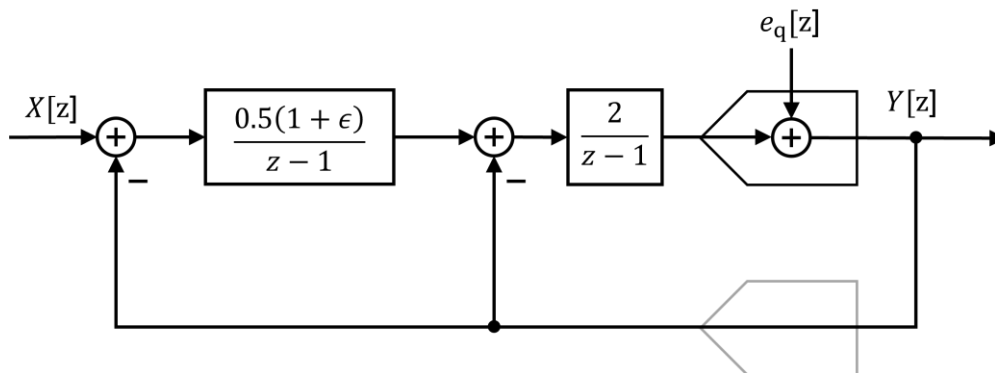


Figure 1: Second-order discrete-time Delta-Sigma ADC.

- a. Show that  $\frac{y_{\text{out}}[z]}{x_{\text{in}}[z]} = 1/(z-1)$  represents a discrete-time integrator.

$$Y_{\text{out}}[n] = Y_{\text{out}}[n-1] + X_{\text{in}}[n-1]$$

**Current output = previous output + previous input = accumulation / integration**

- b. Determine the noise transfer function of the modulator in the z-domain.

$$NTF[z] = \frac{Y[n]}{e_q[n]} = \frac{(z-1)^2}{z^2 + \epsilon}$$

- c. Apply the inverse z-transform, i.e.,  $z = e^{s/f_s} = e^{i2\pi f/f_s}$ , to transfer your result to the s-domain.

$$NTF(s) = \frac{(e^{s/f_s} - 1)^2}{e^{2s/f_s} + \epsilon}$$

- d. Simplify your result by assuming  $f/f_s \ll 1$  and  $e^x \approx 1 + x$  for  $x \ll 1$ .

$$NTF(s) \approx \frac{1}{1 + \epsilon} \left( \frac{s}{f_s} \right)^2$$

e. Which error is modeled by the term  $(1 + \epsilon)$ ?

**gain error**

f. How much does the in-band quantization noise increase for  $\epsilon = -0.05$ ?

**5.3 % or 0.45 dB**

## Task II – Analysis of a Multi-bit ADC / Single-Bit DAC Delta-Sigma ADC

The Delta-Sigma ADC shown in Figure 2 uses a B-bit ADC for increased resolution and a 1-bit DAC to avoid distortion due to DAC nonlinearity. The block *MSB* outputs the most-significant bit of the B-bit ADC.

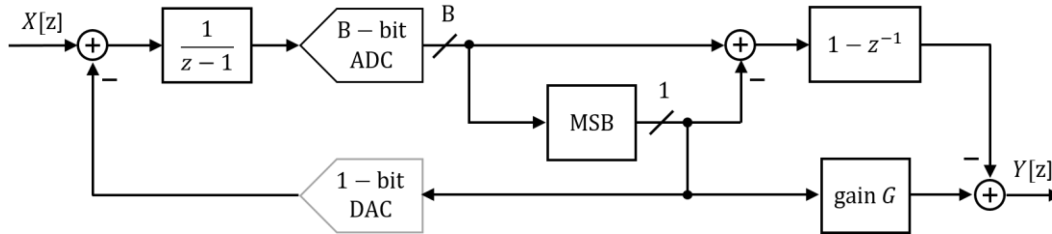
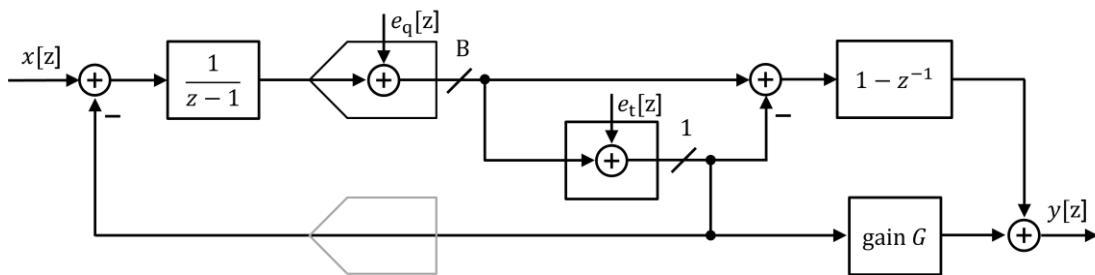


Figure 2: Multi-bit ADC / single-bit DAC Delta-Sigma ADC.

a. Draw the linearized model of the Delta-Sigma ADC assuming that the quantization error  $e_q[z]$  of the B-bit quantizer and the truncation error  $e_t[z]$  of the block *MSB* can be modelled as additive white noise sources.



b. Determine the signal transfer function, the quantization error transfer function, and the truncation error transfer function.

$$STF[z] = G \cdot z^{-1}$$

$$NTF[z] = G \cdot (1 - z^{-1})$$

$$TETF[z] = (G - 1)(1 - z^{-1})$$

c. Considering your result for the truncation error transfer function, what is the optimal gain  $G$ ?

**Set  $G = 1$  to get  $TETF[z] = 0$ .**

- d. Assuming that the gain  $G$  is set to its optimal value, state the  $SQNR$  of this modulator as a function of the resolution  $B$  and the oversampling ratio  $OSR$ .

**As discussed in the lecture**

$$SQNR_p = 6.02 \text{ dB} \cdot B + 10 \log_{10} \left( \frac{3}{2\pi^{2L}} (2L + 1) \right) + (2L + 1) \cdot 10 \log_{10} OSR$$

**For  $L = 1$**

$$SQNR_p = 6.02 \text{ dB} \cdot B - 3.4 \text{ dB} + 30 \log_{10} OSR$$

- e. How much larger is the truncation error compared with the quantization error?

$$e_t \approx 2^B e_q$$

- f. Comment on the potential limitation of this architecture.  
Note: Determine the truncation error transfer function as a function of the noise transfer function (implemented in the analog domain) and the filter function  $1 - z^{-1}$  (implemented in the digital domain).

**$TETF[z]$  is implemented as a mixed-signal transfer function.  $e_t \approx 2^B e_q$  may thus leak to the output if the analog and digital transfer function do not perfectly match.**